

Theorem 1. *For every fixed integer $k \geq 4$,*

$$f_k(n) \gg_k \frac{n^{1-\frac{1}{k-1}}}{(\log n)^{\frac{2(k-2)}{k-1}}}.$$

In particular, the original question has an affirmative answer with

$$c_k = \frac{2(k-2)}{k-1} > 0.$$

Proof. Let $r(k, t)$ denote the least integer N such that every graph on N vertices contains either a copy of K_k or an independent set of size t .

By Theorem 1.1 of Bradač [1], for every fixed $k \geq 3$ there exists a constant $A_k > 0$ such that

$$r(k, t) \geq A_k \frac{t^{k-1}}{(\log t)^{2k-4}}.$$

Set

$$\gamma_k := \frac{2k-4}{k-1} = \frac{2(k-2)}{k-1},$$

and define

$$t_n := \left\lceil B_k n^{1/(k-1)} (\log n)^{\gamma_k} \right\rceil,$$

where $B_k > 0$ is a sufficiently large constant depending only on k .

Since k is fixed,

$$\log t_n = \left(\frac{1}{k-1} + o(1) \right) \log n.$$

Consequently,

$$\frac{t_n^{k-1}}{(\log t_n)^{2k-4}} = \left(B_k^{k-1} (k-1)^{2k-4} + o(1) \right) n.$$

Choose B_k so that

$$A_k B_k^{k-1} (k-1)^{2k-4} > 2.$$

Then, for all sufficiently large n ,

$$r(k, t_n) > n.$$

By the definition of the Ramsey number, there therefore exists an n -vertex graph G containing neither a copy of K_k nor an independent set of size t_n . Thus

$$\alpha(G) < t_n.$$

Every proper coloring of G partitions $V(G)$ into independent sets, and hence

$$n \leq \chi(G) \alpha(G).$$

It follows that

$$\chi(G) > \frac{n}{t_n} \gg_k \frac{n^{1-\frac{1}{k-1}}}{(\log n)^{\frac{2(k-2)}{k-1}}}.$$

Since $f_k(n)$ is the maximum chromatic number among all n -vertex K_k -free graphs, the claimed bound follows. $\square$

References

- [1] D. Bradač, *Off-diagonal Ramsey numbers*, arXiv:2605.28793, <https://arxiv.org/pdf/2605.28793>.